Cloud Native Qumulo Terraform - Complete IAM Analysis

Executive Summary

This document provides a comprehensive analysis of IAM permissions required for deploying and operating the Cloud Native Qumulo (CNQ) infrastructure on AWS using Terraform. The analysis covers both the permissions needed by the Terraform execution role to deploy the infrastructure and the IAM resources created by the Terraform for the deployed infrastructure.

Section 1: IAM Permissions Required for Deployment

These are the permissions that the Terraform execution role (user/role running `terraform apply`) must have to successfully deploy the Cloud Native Qumulo infrastructure.

1.1 Core AWS Service Permissions

EC2 Permissions

json

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "ec2:DescribeInstances",
                "ec2:DescribeInstanceTypes",
                "ec2:DescribeInstanceAttribute",
                "ec2:DescribeImages",
                "ec2:DescribeVpcs",
                "ec2:DescribeSubnets",
                "ec2:DescribeSecurityGroups",
                "ec2:DescribeVolumes",
                "ec2:DescribeSnapshots",
```

```
                "ec2:DescribeKeyPairs",
                "ec2:DescribeAvailabilityZones",
                "ec2:DescribeVpcEndpoints",
                "ec2:RunInstances",
                "ec2:TerminateInstances",
                "ec2:StopInstances",
                "ec2:StartInstances",
                "ec2:RebootInstances",
                "ec2:CreateVolume",
                "ec2:AttachVolume",
                "ec2:DetachVolume",
                "ec2:DeleteVolume",
                "ec2:ModifyInstanceAttribute",
                "ec2:CreateSecurityGroup",
                "ec2:DeleteSecurityGroup",
                "ec2:AuthorizeSecurityGroupIngress",
                "ec2:AuthorizeSecurityGroupEgress",
                "ec2:RevokeSecurityGroupIngress",
                "ec2:RevokeSecurityGroupEgress",
                "ec2:CreateTags",
                "ec2:DeleteTags",
                "ec2:ModifyInstanceMetadataOptions"
            ],
            "Resource": "*"
        }
    ]
}
```

S3 Permissions

json

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
```

```
            "Effect": "Allow",
            "Action": [
                "s3:CreateBucket",
                "s3:DeleteBucket",
                "s3:ListBucket",
                "s3:GetBucketLocation",
                "s3:GetBucketVersioning",
                "s3:GetBucketAcl",
                "s3:GetBucketPolicy",
                "s3:PutBucketPolicy",
                "s3:DeleteBucketPolicy",
                "s3:GetBucketLogging",
                "s3:PutBucketLogging",
                "s3:GetBucketTagging",
                "s3:PutBucketTagging",
                "s3:GetObject",
                "s3:PutObject",
                "s3:DeleteObject",
                "s3:GetObjectVersion",
                "s3:DeleteObjectVersion"
            ],
            "Resource": [
                "arn:aws:s3:::*qumulo*",
                "arn:aws:s3:::*qumulo*/*",
                "arn:aws:s3:::*qps*",
                "arn:aws:s3:::*qps*/*"
            ]
        }
    ]
}
```

IAM Permissions

json

JSON

```json
{
```

```json
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "iam:CreateRole",
                "iam:DeleteRole",
                "iam:GetRole",
                "iam:ListRoles",
                "iam:PassRole",
                "iam:CreateInstanceProfile",
                "iam:DeleteInstanceProfile",
                "iam:GetInstanceProfile",
                "iam:AddRoleToInstanceProfile",
                "iam:RemoveRoleFromInstanceProfile",
                "iam:AttachRolePolicy",
                "iam:DetachRolePolicy",
                "iam:CreatePolicy",
                "iam:DeletePolicy",
                "iam:GetPolicy",
                "iam:GetPolicyVersion",
                "iam:ListPolicyVersions",
                "iam:PutRolePolicy",
                "iam:DeleteRolePolicy",
                "iam:GetRolePolicy",
                "iam:ListRolePolicies",
                "iam:TagRole",
                "iam:UntagRole",
                "iam:ListRoleTags"
            ],
            "Resource": "*"
        }
    ]
}
```

1.2 Additional Service Permissions

Systems Manager (SSM) Permissions

json

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "ssm:GetParameter",
                "ssm:GetParameters",
                "ssm:PutParameter",
                "ssm:DeleteParameter",
                "ssm:DescribeParameters",
                "ssm:GetParameterHistory"
            ],
            "Resource": "arn:aws:ssm:*:*:parameter/qumulo/*"
        }
    ]
}
```

Secrets Manager Permissions

json

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "secretsmanager:CreateSecret",
                "secretsmanager:DeleteSecret",
                "secretsmanager:DescribeSecret",
                "secretsmanager:GetSecretValue",
                "secretsmanager:PutSecretValue",
                "secretsmanager:UpdateSecret",
```

```
                "secretsmanager:TagResource",
                "secretsmanager:UntagResource"
            ],
            "Resource":
"arn:aws:secretsmanager:*:*:secret:qumulo-*"
        }
    ]
}
```

CloudWatch Permissions

json

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "logs:CreateLogGroup",
                "logs:DeleteLogGroup",
                "logs:DescribeLogGroups",
                "logs:PutRetentionPolicy",
                "logs:TagLogGroup",
                "logs:UntagLogGroup",
                "resource-groups:CreateGroup",
                "resource-groups:DeleteGroup",
                "resource-groups:GetGroup",
                "resource-groups:UpdateGroup",
                "resource-groups:Tag",
                "resource-groups:Untag"
            ],
            "Resource": "*"
        }
    ]
}
```

1.3 Optional Service Permissions (Based on Configuration)

Route 53 Resolver Permissions (if `q_cluster_fqdn` and `second_private_subnet_id` are configured)

json

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "route53resolver:CreateResolverEndpoint",
                "route53resolver:DeleteResolverEndpoint",
                "route53resolver:GetResolverEndpoint",
                "route53resolver:ListResolverEndpoints",
                "route53resolver:CreateResolverRule",
                "route53resolver:DeleteResolverRule",
                "route53resolver:GetResolverRule",
                "route53resolver:ListResolverRules",
                "route53resolver:AssociateResolverRule",
                "route53resolver:DisassociateResolverRule",
                "route53resolver:TagResource",
                "route53resolver:UntagResource"
            ],
            "Resource": "*"
        }
    ]
}
```

Elastic Load Balancing Permissions (if `q_nlb_provision = true`)

json

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
```

```
            "Effect": "Allow",
            "Action": [
                "elasticloadbalancing:CreateLoadBalancer",
                "elasticloadbalancing:DeleteLoadBalancer",
                "elasticloadbalancing:DescribeLoadBalancers",

"elasticloadbalancing:DescribeLoadBalancerAttributes",

"elasticloadbalancing:ModifyLoadBalancerAttributes",
                "elasticloadbalancing:CreateTargetGroup",
                "elasticloadbalancing:DeleteTargetGroup",
                "elasticloadbalancing:DescribeTargetGroups",

"elasticloadbalancing:DescribeTargetGroupAttributes",

"elasticloadbalancing:ModifyTargetGroupAttributes",
                "elasticloadbalancing:CreateListener",
                "elasticloadbalancing:DeleteListener",
                "elasticloadbalancing:DescribeListeners",
                "elasticloadbalancing:RegisterTargets",
                "elasticloadbalancing:DeregisterTargets",
                "elasticloadbalancing:DescribeTargetHealth",
                "elasticloadbalancing:AddTags",
                "elasticloadbalancing:RemoveTags"
            ],
            "Resource": "*"
        }
    ]
}
```

KMS Permissions (if `kms_key_id` is specified)

json

```json
{
    "Version": "2012-10-17",
```

```
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "kms:Describe*",
                "kms:List*",
                "kms:Get*",
                "kms:CreateGrant",
                "kms:RevokeGrant"
            ],
            "Resource": "*"
        }
    ]
}
```

1.4 Terraform State Management Permissions

S3 Backend Permissions

json

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "s3:GetObject",
                "s3:PutObject",
                "s3:DeleteObject"
            ],
            "Resource": "arn:aws:s3:::dackss3-dev/tf-state/*"
        },
        {
            "Effect": "Allow",
            "Action": [
```

```
                "s3:ListBucket"
            ],
            "Resource": "arn:aws:s3:::dackss3-dev"
        }
    ]
}
```

Section 2: IAM Permissions and Roles Created by the Infrastructure

This section details the IAM resources that the Terraform creates as part of the Cloud Native Qumulo deployment.

2.1 Qumulo Cluster IAM Role

Role Name Pattern

- `{deployment_unique_name}-qumulo-cluster-iam-role`

Trust Policy

json

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Principal": {
                "Service": "ec2.amazonaws.com"
            },
            "Action": "sts:AssumeRole"
        }
    ]
}
```

Attached Policies

Policy 1: S3 Persistent Storage Access

json

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "s3:GetObject",
                "s3:PutObject",
                "s3:DeleteObject",
                "s3:ListBucket",
                "s3:GetBucketLocation",
                "s3:AbortMultipartUpload",
                "s3:ListMultipartUploadParts",
                "s3:ListBucketMultipartUploads"
            ],
            "Resource": [
                "arn:aws:s3:::*-qps-*",
                "arn:aws:s3:::*-qps-*/*"
            ]
        }
    ]
}
```

Policy 2: EC2 Instance Management

json

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "ec2:DescribeInstances",
                "ec2:DescribeInstanceAttribute",
```

```
                "ec2:DescribeVolumes",
                "ec2:DescribeNetworkInterfaces",
                "ec2:DescribeSecurityGroups",
                "ec2:CreateTags",
                "ec2:ModifyInstanceAttribute",
                "ec2:AssignPrivateIpAddresses",
                "ec2:UnassignPrivateIpAddresses"
            ],
            "Resource": "*"
        }
    ]
}
```

Policy 3: Systems Manager Access

json

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "ssm:GetParameter",
                "ssm:GetParameters",
                "ssm:PutParameter",
                "ssm:DescribeParameters"
            ],
            "Resource": "arn:aws:ssm:*:*:parameter/qumulo/*"
        }
    ]
}
```

Policy 4: CloudWatch Logs (if audit logging enabled)

json

JSON

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "logs:CreateLogStream",
                "logs:PutLogEvents",
                "logs:DescribeLogGroups",
                "logs:DescribeLogStreams"
            ],
            "Resource": "arn:aws:logs:*:*:log-group:qumulo-*"
        }
    ]
}
```

2.2 Qumulo Provisioner IAM Role

Role Name Pattern

- `{deployment_unique_name}-qumulo-provisioner-iam-role`

Trust Policy

json

JSON

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Principal": {
                "Service": "ec2.amazonaws.com"
            },
            "Action": "sts:AssumeRole"
        }
    ]
}
```

Attached Policies

Policy 1: S3 Provisioning Access

json

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "s3:GetObject",
                "s3:PutObject",
                "s3:DeleteObject",
                "s3:ListBucket",
                "s3:GetBucketPolicy",
                "s3:PutBucketPolicy",
                "s3:DeleteBucketPolicy"
            ],
            "Resource": [
                "arn:aws:s3:::dackss3-dev",
                "arn:aws:s3:::dackss3-dev/*",
                "arn:aws:s3:::*-qps-*",
                "arn:aws:s3:::*-qps-*/*"
            ]
        }
    ]
}
```

Policy 2: EC2 Management for Provisioning

json

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
```

```
                "ec2:DescribeInstances",
                "ec2:DescribeInstanceAttribute",
                "ec2:DescribeVolumes",
                "ec2:DescribeNetworkInterfaces",
                "ec2:ModifyInstanceAttribute",
                "ec2:CreateTags",
                "ec2:AssignPrivateIpAddresses",
                "ec2:UnassignPrivateIpAddresses",
                "ec2:StopInstances",
                "ec2:TerminateInstances"
            ],
            "Resource": "*"
        }
    ]
}
```

Policy 3: Secrets Manager Access

json

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "secretsmanager:GetSecretValue",
                "secretsmanager:DescribeSecret"
            ],
            "Resource":
"arn:aws:secretsmanager:*:*:secret:qumulo-*"
        }
    ]
}
```

Policy 4: Systems Manager Access

json

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "ssm:GetParameter",
                "ssm:GetParameters",
                "ssm:PutParameter",
                "ssm:DescribeParameters"
            ],
            "Resource": "arn:aws:ssm:*:*:parameter/qumulo/*"
        }
    ]
}
```

2.3 Instance Profiles

Qumulo Cluster Instance Profile

- Name: `{deployment_unique_name}-qumulo-cluster-instance-profile`
- Associated Role: Qumulo Cluster IAM Role

Qumulo Provisioner Instance Profile

- Name: `{deployment_unique_name}-qumulo-provisioner-instance-profile`
- Associated Role: Qumulo Provisioner IAM Role

2.4 Permissions Boundary Support

The Terraform supports applying IAM Permissions Boundaries to all created roles through the `q_permissions_boundary` variable. When specified:

- All IAM roles created will have the specified permissions boundary policy attached
- This provides an additional security control mechanism
- The permissions boundary policy must already exist in the AWS account
- Example: `q_permissions_boundary = "MyOrgPermissionsBoundary"`

2.5 S3 Bucket Policies (Persistent Storage)

The Terraform creates S3 bucket policies for the persistent storage buckets (when `q_persistent_storage_bucket_policy = true`):

json

JSON

```json
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "QumuloClusterAccess",
            "Effect": "Allow",
            "Principal": {
                "AWS":
"arn:aws:iam::{account-id}:role/{deployment_unique_name}-qumulo-c
luster-iam-role"
            },
            "Action": [
                "s3:GetObject",
                "s3:PutObject",
                "s3:DeleteObject",
                "s3:ListBucket",
                "s3:GetBucketLocation",
                "s3:AbortMultipartUpload",
                "s3:ListMultipartUploadParts",
                "s3:ListBucketMultipartUploads"
            ],
            "Resource": [
                "arn:aws:s3:::bucket-name",
                "arn:aws:s3:::bucket-name/*"
            ]
        },
        {
            "Sid": "QumuloProvisionerAccess",
            "Effect": "Allow",
            "Principal": {
                "AWS":
"arn:aws:iam::{account-id}:role/{deployment_unique_name}-qumulo-p
rovisioner-iam-role"
            },
            "Action": [
                "s3:GetObject",
```

```
                "s3:PutObject",
                "s3:ListBucket",
                "s3:GetBucketPolicy",
                "s3:PutBucketPolicy"
            ],
            "Resource": [
                "arn:aws:s3:::bucket-name",
                "arn:aws:s3:::bucket-name/*"
            ]
        }
    ]
}
```

Section 3: Security Considerations and Best Practices

3.1 Principle of Least Privilege

The IAM roles created by this Terraform follow the principle of least privilege:

- Cluster Role: Limited to operations needed for normal cluster operation
- Provisioner Role: Limited to initial setup and configuration tasks
- Resource-specific permissions: Scoped to Qumulo-related resources where possible

3.2 Resource Scoping

Where possible, permissions are scoped to specific resources:

- S3 permissions limited to Qumulo-specific bucket patterns
- SSM parameters scoped to `/qumulo/*` namespace
- Secrets Manager scoped to `qumulo-*` secret names

3.3 Temporary Resources

The provisioner instance and its associated IAM role are temporary:

- Created during deployment for initial cluster configuration
- Can be terminated after successful deployment
- Should not be relied upon for ongoing operations

3.4 KMS Integration

When using customer-managed KMS keys:

- EBS volumes are encrypted using the specified KMS key

- IAM roles need appropriate KMS permissions for encryption/decryption
- Keys should follow organizational key management policies

3.5 Network Security

The deployment includes security group configurations:

- Cluster security group with configurable CIDR access
- Support for additional security groups via variables
- Default access limited to VPC CIDR block

Section 4: Deployment Recommendations

4.1 Pre-Deployment Checklist

1. IAM Permissions: Ensure Terraform execution role has all required permissions
2. VPC Requirements: Verify S3 Gateway endpoint exists in target VPC
3. Network Configuration: Confirm subnet and security group configurations
4. KMS Keys: If using customer-managed keys, ensure they exist and are accessible
5. S3 Bucket: Ensure the utility S3 bucket exists for provisioning files

4.2 Security Hardening

1. Use Permissions Boundaries: Apply organizational permissions boundaries
2. Enable Termination Protection: Set `term_protection = true` for production
3. Audit Logging: Enable `q_audit_logging = true` for compliance requirements
4. Bucket Policies: Keep `q_persistent_storage_bucket_policy = true` unless managed separately
5. Network Isolation: Use private subnets and configure security groups appropriately

4.3 Monitoring and Maintenance

1. CloudWatch: Monitor created log groups and resource groups
2. IAM Review: Regularly review IAM roles and policies for compliance
3. Security Scans: Include IAM resources in security scanning processes
4. Access Review: Periodically review and rotate credentials

Section 5: Troubleshooting Common IAM Issues

5.1 Deployment Failures

Issue: Terraform fails with IAM permission errors
Solution:

- Review the deployment permissions in Section 1
- Ensure the executing role has all required permissions

- Check for organizational SCPs that might restrict actions

5.2 Cluster Operation Issues

Issue: Qumulo cluster cannot access S3 buckets
Solution:

- Verify the cluster IAM role has correct S3 permissions
- Check S3 bucket policies are correctly applied
- Ensure bucket names match the expected patterns

5.3 Provisioner Issues

Issue: Provisioner fails during cluster setup
Solution:

- Check provisioner IAM role has access to Secrets Manager
- Verify SSM parameter permissions
- Ensure S3 access for provisioning scripts

Conclusion

This IAM analysis provides a comprehensive overview of all permissions required for successful deployment and operation of Cloud Native Qumulo on AWS. The deployment follows AWS security best practices and implements appropriate permission scoping and resource isolation. For any questions or issues related to IAM permissions, refer to the specific sections above or consult the Cloud Native Qumulo documentation.